

# ALCHEMUS

## Advanced Drug Analysis Platform

### Research Owner Wallet

7zZWmnRuKu4KfuUJtvSb7H4uH8PUDq26QK8QY9TG81LJ

Date: 4/30/2025

Compounds: Laughing Gas, LSD

#### RESEARCH PURPOSES ONLY

This analysis is generated for research and educational purposes. The predictions and insights provided should not be considered medical advice.

## New Compound Analysis

Name: Lysergic Nitrous Acetamide

Formula: C<sub>21</sub>H<sub>28</sub>N<sub>2</sub>O<sub>2</sub>

Weight: 340.46

Drug-likeness: 0.65

Activity:

CNS active with psychoactive and anesthetic properties

## Target Analysis

5-HT<sub>2A</sub> Receptor

Binding Affinity: -8.5

Central role in regulating mood, cognition, and perception

NMDA Receptor

Binding Affinity: -7.2

Important for synaptic plasticity and memory function

## Toxicity Profile

Risk Level: Moderate to High

Affected Systems: Brain, Respiratory System

Mitigation Suggestions:

- Careful dose regulation
- Monitoring mental health
- Co-administration with neuroprotective agents

## Pharmacokinetics

Absorption:

Rapid absorption through mucosal membranes

Distribution:

Widely distributed across the CNS

Metabolism:

Hepatic metabolism primarily by CYP2D6 and CYP3A4

Excretion:

Renal; metabolites excreted via urine

Half-life: 6-10 hours

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## Detailed Analysis

Lysergic Nitrous Acetamide is a synthesized compound merging the properties of LSD and Laughing Gas, designed to interact primarily with the serotonergic 5-HT<sub>2A</sub> receptors and the glutamatergic NMDA receptors. This dual receptor modulation results in a unique profile that combines altered sensory perception and mild anesthetic qualities, potentially useful for psychotherapy contexts or as an adjunct in procedural sedation. The compound's drug-likeness is moderately high but balanced by cautionary toxicity risks, primarily due to its psychoactive nature and impact on the CNS. With a comprehensive pharmacokinetic profile indicating rapid absorption and a natural metabolic pathway, this compound represents an intriguing novel agent with the potential to bridge psychoactivity and anesthesia. Ongoing research is aimed at optimizing its safety profile and therapeutic effectiveness.

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Combine base compounds, simulate new drug possibilities with AI, and earn rewards through Solana blockchain. Join our community of researchers and innovators in shaping the future of pharmaceutical discovery.

<https://ALCHEMUS.ai>